# MILITARY SPECIFICATION

# FOR

# CONVERTER, RADAR DATA

# CV-1607D/AVA-1

## 1. SCOPE

1.1 Scope. This specification covers the design, performance, and acceptance requirements for the Radar Data Converter (RDC) CV-1607D/AVA-1 herein referred to as the RDC. The RDC generates a shade of gray terrain clearance presentation which is displayed on the Analog Display Indicator IP-722( )/AVA-1.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, Standards, and Handbooks. Unless otherwise specified, the following specifications, and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-B-5087 | Bonding, Electrical, and Lightning Protection for Aerospace Systems |
| MIL-W-5088 | Wiring, Aircraft, Installation of |
| MIL-E-5400 | Electronic Equipment, Airborne, General Specification for |
| MIL-T-5422 | Testing, Environmental, Airborne Electronic and Associated Equipment |
| MIL-M-7793 | Elapsed Time Meters |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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|---|---|
| MIL-E-17555 | Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of |
| MIL-D-18300 | Design Data Requirements for Avionic Equipment |
| MIL-T-18303 | Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for |
| MIL-N-18307 | Nomenclature and Identification for Electronic, Aeronautical and Aeronautical Support Equipment Including Ground Support Equipment |
| MIL-M-38510 | Microcircuits, General Specifications for |
| MIL-I-83287 | Indicators, Fault Locating, General Specification for |
| MIL-S-19500 | Semiconductor Devices, General Specification for |

NAVAL AIR SYSTEMS COMMAND

| | |
|---|---|
| AS-4613 | Application and Derating Requirements for Electronic Components, General Specification for |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-461 | Electro-Magnetic Interference Characteristics Requirements for Equipment |
| MIL-STD-462 | Electro-Magnetic Interference Characteristics, Measurement of |
| MIL-STD-470 | Maintainability Program for Systems and Equipment |
| MIL-STD-471 | Maintainability Demonstration |
| MIL-STD-704 | Aircraft Electric Power Characteristics |
| MIL-STD-781 | Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution |
| MIL-STD-785 | Reliability Program for Systems and Equipment Development and Production |
| MIL-STD-794 | Parts and Equipment, Procedures for Packaging and Packing of |
| MIL-STD-810 | Environmental Test Methods |



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| MIL-STD-882 | System Safety Program for Systems and Associated Subsystems and Equipment; Requirements for |
| MIL-STD-965 | Parts Control Program |
| MIL-STD-2074 | Failure Classification for Reliability Testing |
| MIL-STD-2084 | General Requirements For Maintainability of Avionic and Electronic Systems and Equipment |
| MS 17322 | Meter, Time Totalizing, Miniature Digital 115V, 400 Cycle |

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (See 4.4 and 6.3).

3.2 Parts, materials, and processes. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall be in accordance with the requirements specified in MIL-M-38510 and MIL-S-19500.

b. Other parts and materials requirements shall be in accordance with MIL-E-5400.

c. Nonrepairable subassemblies shall be used in accordance with MIL-E-5400.

d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Parts derating. All electronic components shall meet the application and derating requirements of AS-4613 for Class A equipment.



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3.2.2 Nonstandard parts and material approval. Approval for use of nonstandard parts and materials other than microelectronic devices shall be obtained as specified in MIL-E-5400. Microelectronic devices shall be approved as specified in MIL-STD-965.

3.2.3 Limited life items. The RDC shall not contain limited life items which require scheduled replacement within the minimum specified operating life of the RDC.

3.2.4 Microelectronic modular assemblies. When used, microelectronic modular assemblies shall meet the requirements of MIL-STD-965.

3.2.5 Modules. The electronic portions of the RDC shall be functionally modularized in accordance with MIL-STD-2084. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly. Modules shall not be hardwired.

3.3 Design and construction. Unless otherwise specified herein, the RDC shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship. Design data shall be prepared in accordance with MIL-D-18300 and made available to the procuring activity. (See 6.2.2)

3.3.1 Weight. The weight of the RDC, excluding cables, shall be a minimum consistent with good design and shall be not greater than 25 pounds (11.38 Kg). The total weight shall include provisions for mounting.

3.3.2 Configuration. The dimensions of the RDC and external hardware locations shall be in accordance with Figure 1.

3.3.2.1 Mounting. Mounting provisions for the RDC shall be four 10-32 UNF-3A captive screws.

3.3.2.2 Time totalizing meter. The RDC shall contain a time totalizing meter in accordance with MIL-M-7793 and MS 17322.

3.3.2.3 Connectors. The interface connectors shown in Figure 1 shall be as specified below:

a. J1-Deutsch DS00-3719-PW-059
b. J2-Deutsch DS00-3719-PX
c. J3-Amphenol 67-00P20-37P
d. J4-Deutsch DBA-70-37-0SN
e. J5-Deutsch DS00-61-5S

J4 and J5 shall contain provisions for captive connector caps.



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NOTE: Dimensions without tolerance are for reference only. Tolerances shall be as specified on applicable drawings.

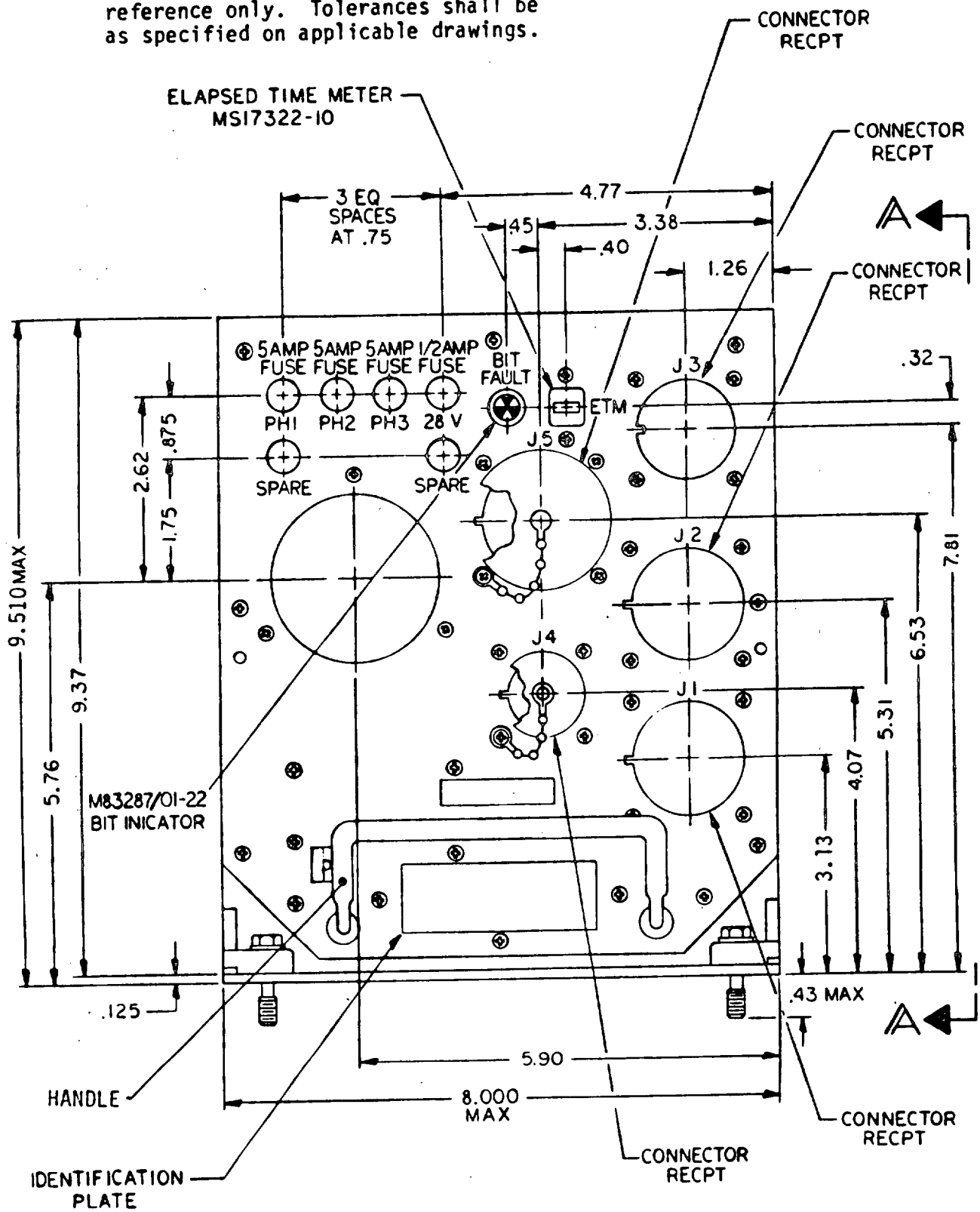



FIGURE 1. Radar data converter.



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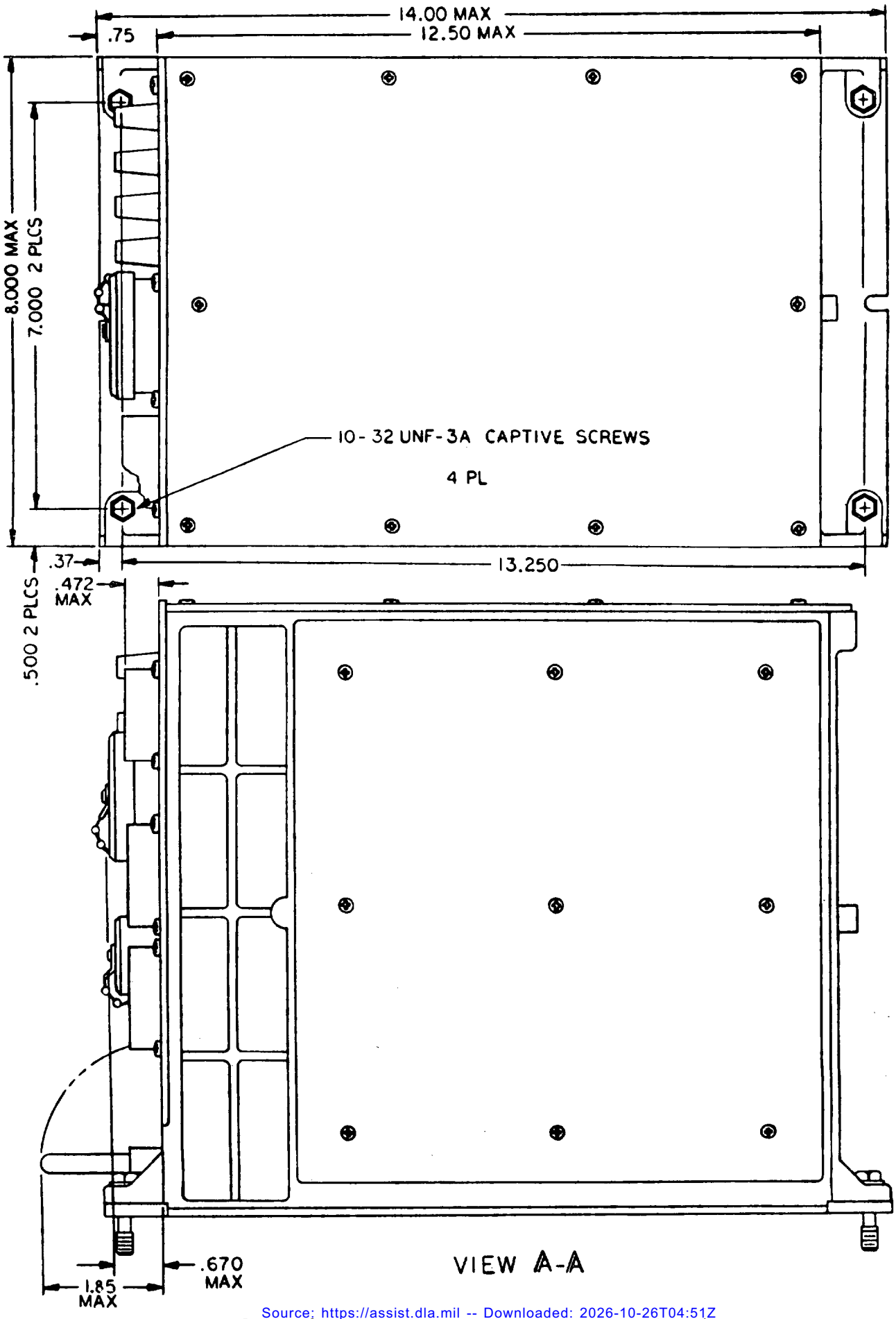



FIGURE 1. Radar data converter - Continued.

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3.3.2.4 Fuses. The RDC shall provide circuit protection to the input power by means of fuses. A spare fuse for the 115 volt power and a spare fuse for the 28V power shall be provided.

3.3.2.5 BIT indicator. The RDC shall contain a BIT indicator in accordance with MIL-I-83287/01-22.

3.3.3 Reliability. The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a reorder from a contractor who is producing the RDC, the program previously approved may be continued unless otherwise indicated in the contract or order.

3.3.3.1 Operational stability. The RDC shall operate continuously or intermittently with satisfactory performance, without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.3.2 Operating life. The RDC shall have a total operating life of 10,000 hours with minimal servicing and replacement of parts.

3.3.3.3 Mean Time Between Failure (MTBF). The RDC, including all Built-In-Test (BIT) provisions, shall have an MTBF of 1200 operating hours.

3.3.3.4 Storage service life. At any time within, or immediately following an 18 month storage period under any combinations of climatic environments specified herein, the RDC shall be capable of meeting all requirements of this specification without component or part replacement, readjustment, or other maintenance actions.

3.3.4 Cabling and connections.

3.3.4.1 Cables and connectors. The RDC shall provide for the use of wiring and connectors in accordance with MIL-E-5400.

3.3.4.2 Interconnection cabling. The RDC shall be capable of specified operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the RDC.

3.3.5 Interchangeability. The RDC shall meet the interchangeability requirements of MIL-E-5400.

3.3.6 Maintainability. The maintainability program and maintainability requirements shall be in accordance with MIL-STD-470 and MIL-STD-2084 and with the additional requirements specified herein.

3.3.6.1 Maintenance requirements. The maintenance plan for the RDC shall include the requirements specified in 3.3.6.1.1 through 3.3.6.1.3.

3.3.6.1.1 Preventive maintenance. The RDC shall not have preventive maintenance requirements.



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3.3.6.1.2 Organizational level. The BIT function shall be used for fault isolation to the Weapon Replaceable Assembly (WRA). This level of maintenance will remove the defective WRA and replace it with a Ready For Issue (RFI) WRA. The WRA will be tested after replacement. No Ground Support Equipment (GSE) shall be required.

a. BIT shall be incorporated into the RDC in accordance with 3.3.6.2.2.

b. Removal and replacement of the RDC shall be not greater than 12 minutes mean-time (exclusive of access time).

c. The total organizational level corrective maintenance time, Turn Around Time (TAT), shall be the sum of time required for fault isolation, removal and replacement, and repair verification. The TAT shall be not greater than 20 minutes mean-time.

3.3.6.1.3 Intermediate level. Intermediate level maintenance generally shall consist of the repair of WRAs and Shop Replaceable Assemblies (SRA) in off-aircraft shops. The Quick Replaceable Assembly (QRA) ratio shall be not less than 0.90.

a. WRA repair - The RDC shall be compatible with Mini-Shop Automatic Checkout Equipment (Mini-SACE) via the RDC adapter drawer for fault detection, isolation to the SRA, removal and replacement of the SRA, and WRA operational verification.

b. SRA repair - SRAs shall have test points available which allow fault detection and isolation to the faulty piece part group. Repair is achieved by the removal and replacement of the identified piece part(s). SRA operational verification is accomplished by test point examination and by operation of the SRA in the RDC.

c. Total intermediate level corrective maintenance - The summation of removal and replacement of defective SRA and the removal and replacement of a defective part(s) including fault isolation, access, and secure time, shall not exceed 92 minutes mean-time.

d. Mini-SACE compatibility - RDC compatibility with Mini-SACE (AN/ASM-461) shall be demonstrated. The RDC shall be compatible with Mini-SACE to the extent that the Mini-SACE interface presented to the RDC is mechanically identical and to the extent that the Mini-SACE functional interface parameters have a common operating band.

3.3.6.1.4 Depot level. Depot level maintenance shall not be required.

3.3.6.2 General maintainability requirements.

3.3.6.2.1 Task criteria. Each of the maintenance times specified in 3.3.6.1.2 shall be accomplished with a maximum of two maintenance personnel of no higher than intermediate skill (E-5) for at least 95 percent of the maintenance tasks.



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3.3.6.2.2 Built in test. Suitable BIT features shall be incorporated into the RDC to provide the operator with indications of RDC readiness and to locate the fault to the malfunctioning WRA. These features shall be implemented using automatic techniques as well as manually activated test pattern(s) displayed on the Analog Display Indicator (ADI) and shall be capable of unambiguously isolating to the defective WRA for at least 95 percent of all possible RDC faults when the faults are ranked in order of their occurrence with a 90 percent confidence level.

3.3.6.2.2.1 Test display. During the test mode, the RDC shall provide a static test display. The test display shall exercise the complete capabilities of the RDC and unambiguously isolate to the WRA. The test display shall be easily interpreted by the operator and shall not require complex mental calculations. The test display shall be coordinated with and approved by the procuring activity.

3.3.6.2.2.2 BIT indicator. A latching BIT indicator shall be provided which retains the WRA BIT status in conditions of power off.

3.3.6.2.3 Test points. Test points shall satisfy the following requirements:

a. Functional test point arrangement - Test points shall be exposed and grouped within multipin connectors which are readily accessible at the level of maintenance for which they are intended. The test points shall be grouped by the functional maintenance level requirements as follows:

(1) WRA test points for intermediate level maintenance.

(2) SRA test points for intermediate level maintenance.

b. The test points shall be located as follows:

(1) WRA test points shall be located on exterior surfaces of the WRA.

(2) SRA test points shall be located in such positions so that they may be readily accessible for maintenance actions.

3.3.6.2.4 Shop replaceable assembly installation. SRAs shall be installed in the RDC such that, when the RDC is mounted in the aircraft, intermittent operation due to catapult launches and arrested recoveries is precluded.

3.3.6.2.5 Extractors. SRAs shall contain attached extractors, to simplify SRA insertion, withdrawal and replacement, which utilize mechanical leverage. It shall not be necessary to restrain the WRA chassis in order to remove or install SRAs. The extractors shall also provide a positive locking device for the SRAs to assure proper seating of the connector pins and to restrain the SRA in the WRA. SRAs shall not be restrained by resilient pads on access covers.



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3.3.7 Surrounding air pressure. For design purposes air pressure surrounding the RDC shall be assumed equivalent to free stream static pressure. All newly developed RDCs shall be designed to satisfactorily withstand or operate at 1.33 psia surrounding air pressure.

3.3.7.1 Normal. The surrounding air pressure may vary at a rate as high as 0.25 psi per second between 21.0 and 2.73 psia.

3.3.8 Thermal requirements and cooling. The RDC shall meet the thermal design requirements specified in MIL-E-5400.

3.3.8.1 Thermal design. The RDC shall not require cooling by means other than conduction through the mounting hardware and convection into the equipment bay environment. In the event that air vents are required, they shall be protected as necessary to preclude the entry of foreign material. A fan, integral with the RDC may be provided to maximize convection cooling.

3.3.8.2 Heat dissipation. The heat dissipation for the RDC shall be not greater than 170 watts, exclusive of cooling fan.

3.3.8.3 Exterior touch temperature. The exterior temperature of the RDC shall be not greater than 35°C over the ambient temperature.

3.3.9 Human factors engineering. The requirements shall conform to MIL-E-5400.

3.3.10 Bonding. The RDC shall be bonded in accordance with MIL-B-5087.

3.3.11 Safety. The RDC shall be designed to preclude the incorporation of features which result in critical or catastrophic hazards as classified in MIL-STD-882. Personnel safety shall be in accordance with MIL-E-5400.

3.4 Performance. Unless otherwise specified herein, values set forth to establish the requirements of specified performance apply to performance under both standard and extreme service and all input power conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions shall be specified.

3.4.1 Service conditions - electrical.

3.4.1.1 Primary input power requirements. Except as specified herein, the RDC shall meet all applicable requirements of MIL-E-5400 and shall give specified performance from the following power sources having characteristics as defined in MIL-STD-704. The maximum steady state power required shall be not greater than the limits specified below:

a. AC power: (Three phase) 115/200V, 400 Hz Category B, 200 VA maximum.

b. DC power: 28V, 25 watts maximum.

Power shall be externally switched such that power shall be provided when the RDC is to operate and the voltage shall be zero when the RDC is to be non-operating.



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3.4.1.2 Steady-state and transient operations. The RDC shall provide specified performance when the prime power inputs are within the full performance limits specified in MIL-STD-704. Under transients within limits of MIL-STD-704 the RDC shall remain safe, shall automatically recover to full performance, and shall remain unaffected in reliability.

3.4.1.3 Overload protection. Overload protection shall be provided in the RDC. All parts and circuits which are likely to carry an overload due to any failure or poor adjustments shall have suitable overload protective devices or shall be proportioned to withstand such overload without permanent damage to the RDC. The use of fuses or other protective devices shall be held to a minimum.

3.4.1.4 Under voltage protection. The RDC shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.4.1.5 Loss of power protection. A 400 Hz single phase or a balance loading of Y connected three phase shall be used. The RDC shall not suffer damage nor shall it cause heavy loading on the remaining phases, if one of the phase leads does not supply power to the RDC.

3.4.2 Warm-up time. The time required for the RDC to warm up prior to operation shall be kept to a minimum and shall be not greater than two minutes under standard conditions and at extreme service conditions.

3.4.3 Acoustic noise generation. The noise level generated by the RDC shall be not greater than 85 decibels (referenced to 0.002 dynes per square centimeter) measured at any point two feet from the RDC.

3.4.4 Shaded gray terrain clearance presentation. The RDC shall generate and provide Terrain Clearance (TC) video for display on the Analog Display Indicator (ADI).

3.4.4.1 TC video. The RDC shall generate video which can be displayed in a raster format. The video shall consist of the following symbols which shall conform to the detail requirements of 3.5.

a. Eight Range Contours
b. Highlight Bars
c. Black Altitude Curtain
d. Artificial Horizon Line
e. Sky
f. Ground
g. Tactical Altitude Director (only if activated)
h. Calibration Markers

3.4.4.2 Pitch destabilization. Pitch destabilization shall be continuous between ±30 degrees of elevation.

3.4.4.3 Lag compensation. A programmable lag adjustment shall be provided to shift the TC presentation in order to offset the TC phase video - antenna position lag. The processed terrain clearance video shall be shifted in azimuth by 0.255 degree increments.



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3.4.4.5 Search radar scan angle variation. When the search radar azimuth scan varies from ±30 degrees to ±60 degrees, the distribution of azimuth elements shall remain as specified in 3.5.2.6.

3.4.4.6 TCAV absence. The RDC shall provide failsafe video of a nontransparent range 1 contour to the artifical horizon line in the event that the TCAV absence signal (see 3.4.8.2.5) indicates unsatisfactory radar returns.

3.4.5 TC calibrate. The RDC shall generate calibration markers at the right side of the video when TC CAL off/on signal is on. The calibration markers shall conform to the detail requirements of 3.5.5.

3.4.6 Test mode. The RDC shall generate a test mode video which shall contain all symbols including the eight range contours. The test mode shall be selected by the test mode on/off signal from the pilot's control panel. The calibration markers shall not be generated.

3.4.7 Failure detection. The RDC shall detect the loss of the following input signals:

a. TCP to RDC trigger
b. Azimuth scan
c. Vertical sync
d. Horizontal sync
e. Processed TC video (loss of calibration shall also be detected)
f. Azimuth angle

The RDC validity output shall be 28 VDC if any of these failures occur. (The 28 VDC signal is routed through the search radar to the ADI and sets the RDC indicator light to invalid.) For failures a or b, the range 1 contour shall also be positioned at the artificial horizon line.

3.4.7.1 BIT failure detection. Internal failure of the RDC shall cause the RDC validity output to be 28 VDC and shall set the BIT indicator to its fail position.

3.4.8 Inputs. The following inputs (see Table I) shall be supplied to the RDC. The RDC interface with associated equipment is shown in Figure 2.

3.4.8.1 Inputs from the inertial navigation system.

3.4.8.1.1 Elevation angle. The elevation angle shall have a scale factor of 3 degrees per volt with a positive voltage for positive angles.

3.4.8.1.2 Azimuth angle. The azimuth angle shall be 0.3 to 0.8 V peak to peak 31.5 kilohertz sine wave which shifts phase relative to horizontal sync.

3.4.8.2 Inputs from the search radar.

3.4.8.2.1 Processed TC video. Processed TC video shall be an analog of terrain elevation angle versus slant range, where terrain elevation angle is measured in volts/degree and range in nautical miles/microsecond. Terrain elevation angle shall vary between the limits of +8 degrees and -32 degrees.



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TABLE I. Input signal characteristics

| Input number | Signal name | Function | Signal type | Voltage range | Zin-input impedance | RDC connector pin | Signal source |
|---|---|---|---|---|---|---|---|
| 1 | Processed terrain clearance video | Radar video | Analog | See figure 3 | 1K | J1-10 (Coax) | Search radar |
| 2 | Azimuth scan | Antenna azimuth scan angle | Analog ramp | ±3V (∓ 30°) or ±6V (∓ 60°) | 200K ohm | J2-14 | Search radar |
| 3 | Azimuth scan shield | Shield termination | - | - | - | J2-1 | Search radar |
| 4 | Scan angle limit | Antenna scan direction | Square wave | 28 (+4) VDC (-11) =Scan right 0 (+3) VDC (-0) =Scan left | 20K ohm | J3-D | Search radar |
| 5 | TCP to RDC trigger | Timing synchronization | Positive pulse | +20 ± 5 volt pulse 2.0 ± 1 usec duration | 180 ohm | J1-8 (Coax) | Search radar |
| 6 | 280 usec trigger | Timing synchronization | Positive pulse | +20 ± 5 volts pulse 2.0 ± 1 usec duration | 180 ohm | J1-12 (Coax) | Search radar |





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TABLE I. Input signal characteristics - Continued

| Input number | Signal name | Function | Signal type | Voltage range | Zin-input impedance | RDC connector pin | Signal source |
|---|---|---|---|---|---|---|---|
| 7 | TCP/MOD trigger | Timing synchroniz-ation | Positive pulse | +20 ± 5 volt pulse 2.0 ± 1 usec duration | 91 ohm | J1-11 (Coax) | Search radar |
| 8 | TCAV absence | Display of range 1 video at 0 degree elevation | Discrete | +28 V=absent 0 VDC or open=normal | 2K ohm | J2-7 | Search radar |
| 9 | Horizontal sync | Output Timing synchroniza-tion | Positive pulse | 5 ± 0.5 volts, 15.75 KHZ | 220 ohm | J2-11 (Coax) | ADI |
| 10 | Vertical sync | Output Timing Synchroniza-tion | Positive pulse | 5 ± 0.5 volts, 60 HZ | 330 ohm | J2-9 (Coax) | ADI |
| 11 | Test mode on/off | Test mode on/off | Discrete | +28 VDC= test mode 0 VDC or Open=normal | 2K ohm | J3-T | Pilot's control panel |
| 12 | TC Cal off/on | Off/on for calibration display | Discrete | +28 VDC=off, 0 VDC or Open=on | 2K ohm | J1-1 | Pilot's control panel |





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TABLE I. Input signal characteristics - Continued

| Input number | Signal name | Function | Signal type | Voltage range | Zin-input impedance | RDC connector pin | Signal source |
|---|---|---|---|---|---|---|---|
| 13 | 4/6 range highlight select | Coded 2 wire selection of highlighted range | Discrete | 0 VDC or open = not selected, 28 VDC = selected | 1K ohm | J2-6 | Pilot's control panel |
| 14 | 5/6 range highlight select | Same as 13 | Discrete | Same as 13 | 1K ohm | J2-18 | Pilot's control panel |
| 15 | Pilot's control panel signal ground | Ground return | - | - | - | J2-5 | Pilot's control panel |
| 16 | Altitude above terrain | Radar altitude positions black altitude curtain | Analog,DC | +1 to +25 VDC | 100K | J1-4 | Radar altimeter |
| 17 | Altitude above terrain return | Ground return | - | - | - | J1-5 | Radar altimeter |





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TABLE I. Input signal characteristics - Continued

| Input number | Signal name | Function | Signal type | Voltage range | Zin-input impedance | RDC connector pin | Signal source |
|---|---|---|---|---|---|---|---|
| 18 | Altitude above terrain shield | Shield termination | - | - | - | J1-7 | Radar altimeter |
| 19 | 115 V, 400 Hz phase 1 | Switched primary power | 115 V, 3 phase 400 Hz | MIL-STD-704 | - | J3-a | Pilot's control panel |
| 20 | 115 V, 400 Hz phase 2 | - | - | MIL-STD-704 | - | J3-b | Pilot's control panel |
| 21 | 115 V, 400 Hz phase 3 | - | - | MIL-STD-704 | - | J3-c | Pilot's control panel |
| 22 | Elevation angle | Aircraft pitch | Analog,DC | ± 30 VDC | 100K | J3-j | INS |
| 23 | Azimuth angle | Aircraft heading | Resolver | 0.3 to 0.8 VAC pp 31.5K Hz | 9.4K to 12K (AC coupled) | J3-m | INS |
| 24 | Elevation angle return | Ground return | - | - | - | J3-i | INS |





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TABLE I. Input signal characteristics - Continued

| Input number | Signal name | Function | Signal type | Voltage range | Zin-input impedance | RDC connector pin | Signal source |
|---|---|---|---|---|---|---|---|
| 25 | Signal ground | Ground return | - | - | - | J1-6 | Search radar |
| 26 | Search radar azimuth scan return | Ground return | - | - | - | J2-2 | Search radar |
| 27 | +28 VDC power | Aircraft power | DC | MIL-STD-704 | - | J3-p | Aircraft |
| 28 | +28 VDC return | Power return | - | - | - | J3-n | |
| 29 | AC neutral | Power return | - | - | - | J3-h | |
| 30 | AC power shield | Shield termination | - | - | - | J3-q | Aircraft |
| 31 | Impact point | Horizontal position | DC | -15 to +15 | 10K | J3-H | ADI |
| 32 | Impact point | Vertical position | DC | -15 to +15 | 10K | J3-J | ADI |





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TABLE I. Input signal characteristics-Continued

| Input number | Signal name | Function | Signal type | Voltage range | Zin-input impedance | RDC connector pin | Signal source |
|---|---|---|---|---|---|---|---|
| 33 | Search radar | On/off | DC | 28/0 | 2K | J3-N | ADI |
| 34 | Land gear | Up/down | DC | Open or 0/28 | 2K | J3-R | ADI |
| 35 | ADI TAD | TAD/ no TAD | DC | 0V = TAD<br>5V = no TAD | - | J3-S | ADI |
| 36 | Computer gnd | Impact point signal return | DC | - | - | J3-K | ADI |





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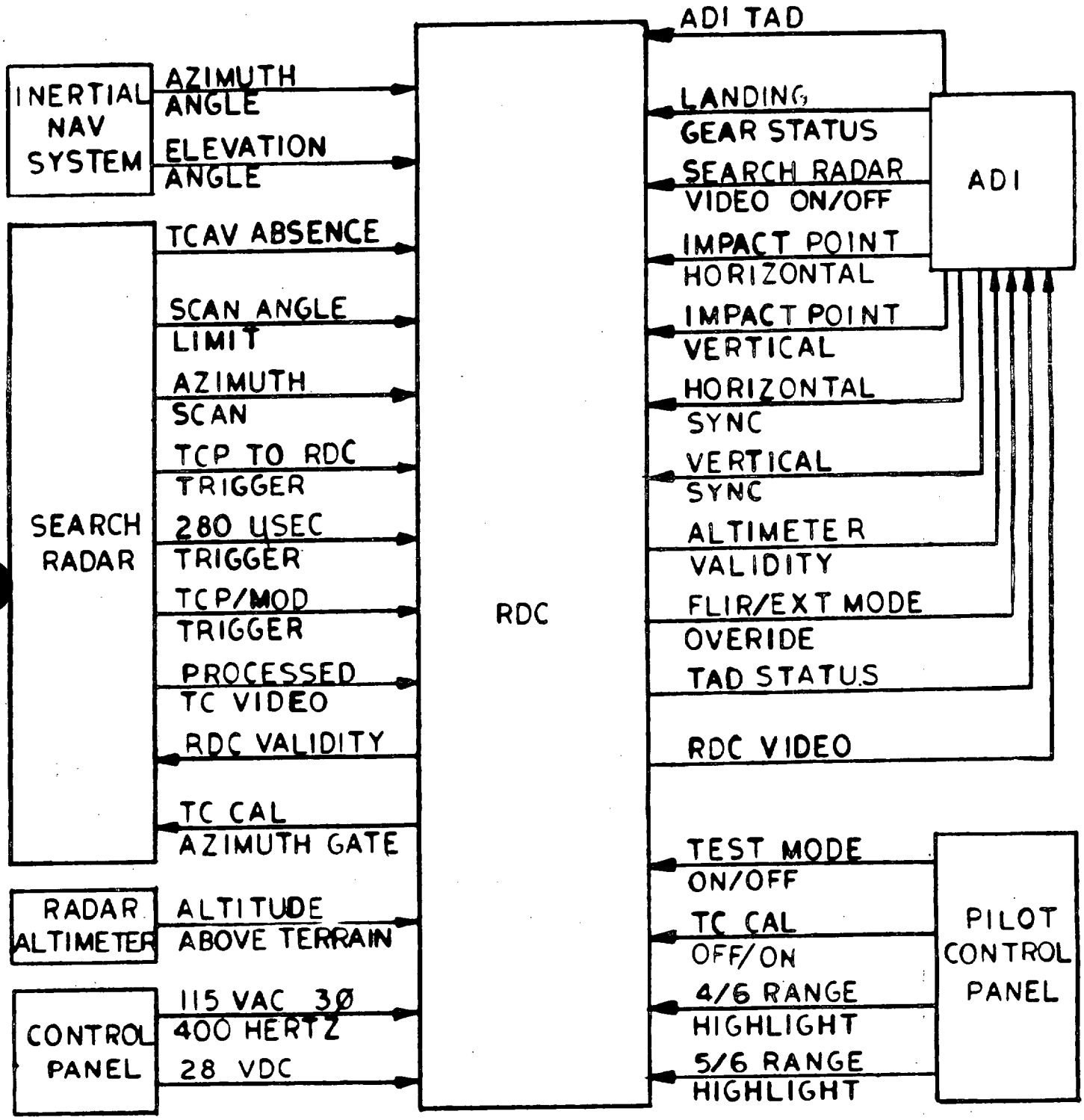



FIGURE 2. RDC system interface.



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The signal input shall be terminated in a 1000 ohm impedance. The characteristics of the signal are shown in Figure 3. The video shall be clamped in the RDC to the zero degree (horizon) reference.

3.4.8.2.2 Scan angle limit. This signal shall consist of two discrete levels: +28 (+4, -11) VDC when the antenna is scanning to the right and 0 (+3, -0) VDC when the antenna is scanning to the left. The change in voltage level will occur approximately 2.5 degrees before the antenna arrives at its limit. This signal shall be terminated in 20K ohms or greater.

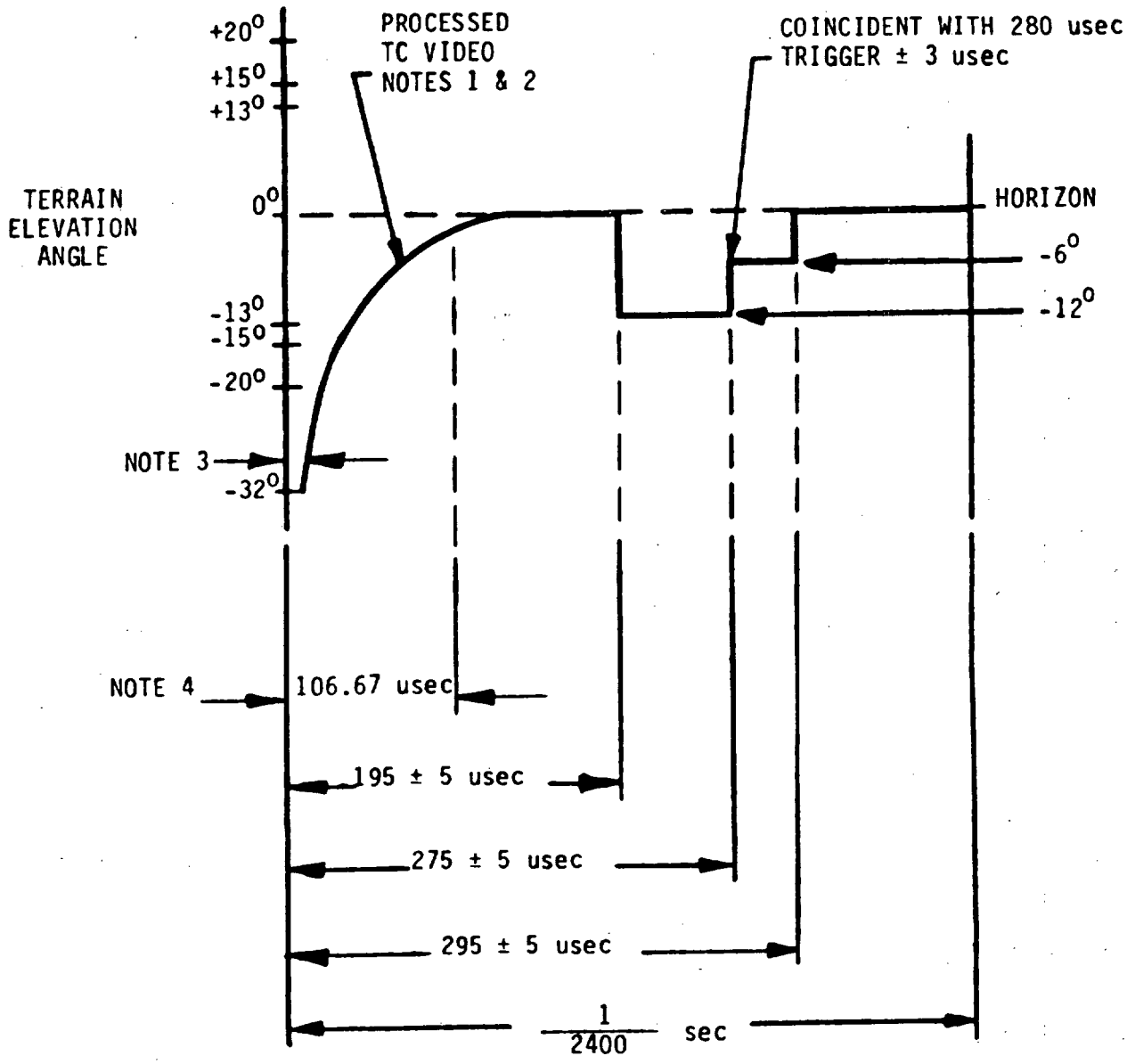
3.4.8.2.3 Azimuth scan. An antenna scan angle sweep voltage shall be available from the search radar with the requirements specified in Table II.

TABLE II. Azimuth scan requirements.

| Parameter | Requirements |
|---|---|
| Scale factor | -10 degree/volt |
| Sense | +3.0 volts corresponding to -30 degrees left of aircraft centerline.<br>-3.0 volts corresponding to +30 degrees right of the aircraft centerline. |
| Zero offset | ±50 mv |
| Source impedance | 12K ohms |
| Load impedance (RDC) | 200K ohms or greater |
| Noise | Less than 20 mv ripple, less than 200 mv spikes due to pulse kickbacks. |
| Scan angle variation | ±30 to ±60 degrees |
| Scan rate | 100 degrees per second |



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NOTES:

1. Shape and slope changes in accordance with Terrain and aircraft altitude. Scale factor is 237.33 millivolts per degree.
2. When Terrain exceeds horizon, waveform peaks at higher level (Above 0°).
3. Width varies as a function of altitude.
4. Equivalent to range of Apprx 8.5 nautical miles.

FIGURE 3. Processed TC video.



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3.4.8.2.4 Timing pulses. The radar timing pulses specified in 3.4.8.2.4.1 through 3.4.8.2.4.3 shall be provided. Timing relationships shall be as shown in Figure 4. Pulse characteristics are as specified in Table III.

TABLE III. Timing pulses.

| Signal Parameter | TCP/MOD Trigger | 280 usec Trigger | TCP to RDC Trigger |
|---|---|---|---|
| Amplitude | 20 ± 5 volts | 20 ± 5 volts | 20 ± 5 volts |
| Pulse width | 2 ± 1 usec | 2 ± 1 usec | 2 ± 1 usec |
| Rise time | 0.20 usec max | 0.20 usec max | 0.20 usec max |
| Fall time | 0.20 usec max | 0.20 usec max | 0.20 usec max |
| Termination | 91 ohm | 180 ohm | 180 ohm |
| PRF | 2000 to 2430 Hz | 2000 to 2430 Hz | 2000 to 2430 Hz |
| Polarity | Positive | Positive | Positive |

3.4.8.2.4.1 TCP to RDC trigger. A positive search radar TCP to RDC trigger pulse shall be provided at PRF of 2400 Hz. This pulse shall be used to trigger TC range sampling circuitry. The trigger timing relationship occurs in two variants, which are externally controlled. These result in two unique displays. The variants are:

a. Coincident with TCP/MOD trigger. (normal mode)

b. Coincident with 280 usec trigger. (calibration mode)

3.4.8.2.4.2 TCP/MOD trigger. A positive trigger pulse from the modulator to the terrain clearance processor shall also be provided to the RDC. This signal is the zero time reference for the processed TC video.

3.4.8.2.4.3 280 microsecond trigger. This signal shall be provided 280 microseconds after TCP/MOD trigger.

3.4.8.2.5 TCAV absence. This discrete signal indicates absence of valid processed TC video. When this signal is present, range one contour shall be positioned at the artificial horizon line. When the TC cal off/on signal is also on, the calibration markers shall be displayed but their position will not necessarily be valid.

3.4.8.3 Inputs from the ADI. The inputs of 3.4.8.3.2, through 3.4.8.3.6 may not be present if the aircraft or the ADI have not been modified.

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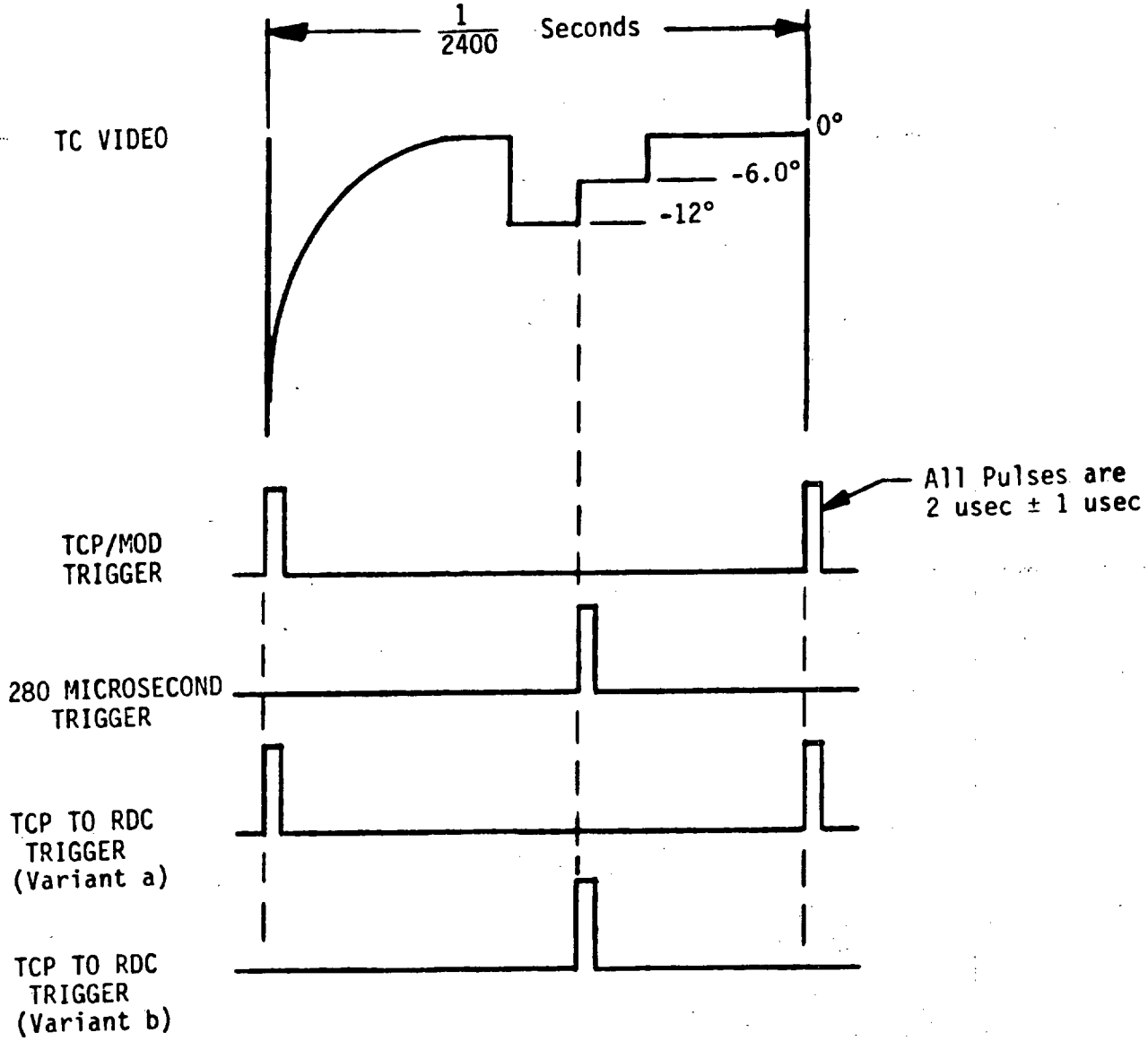



FIGURE 4. Search radar timing pulses.



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3.4.8.3.1 Horizontal sync and vertical sync. The ADI shall provide horizontal and vertical sync pulses to the RDC. Amplitudes and tolerances are as specified in Table IV.

TABLE IV. Horizontal sync and vertical sync.

| Characteristics | Horizontal sync pulse | Vertical sync pulse |
|---|---|---|
| Amplitude | 5 volts ± 10 percent | 5 volts ± 10 percent |
| Pulse width | 12 usec ± 1.0 usec | 500 usec ± 50 usec |
| Rise time | 1.0 usec maximum | 3 usec maximum |
| Fall time | 1.0 usec maximum | 12 usec maximum |
| Termination | 220 ohms direct coupled to ground | 330 ohms direct coupled to ground |
| Polarity | Positive | Positive |
| PRF | 15,750 Hz | 60 Hz |

3.4.8.3.2 Impact point horizontal position. An analog DC voltage (-15 to +15V) shall be provided which shall be proportional to the horizontal displacement in roll stabilized aircraft coordinates of the present flight path direction. A positive signal shall indicate displacement to the left; a negative signal shall indicate a displacement to the right.

3.4.8.3.3 Impact point vertical position. An analog DC voltage (-15 to +15 V) shall be provided which shall be proportional to the vertical displacement of the present flight path direction. A positive signal shall indicate a displacement downward; a negative signal shall indicate a displacement upward.

3.4.8.3.4 Search radar video on/off. A DC voltage shall be supplied to indicate whether the ADI is in TC or contact analog mode. (28 VDC for TC)

3.4.8.3.5 Landing gear status. A DC voltage (+28 VDC for down and 0 VDC or open for up) shall indicate the status of the aircraft landing gear.

3.4.8.3.6 ADI TAD. Zero volts DC shall indicate the ADI and the aircraft have been modified for TAD. An open shall indicate the ADI or the aircraft has not been modified for TAD.



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3.4.8.4 Inputs from the pilot's control panel. The following discrete +28 VDC inputs shall be provided by the pilot's control panel:

a. Test on-off: +28 VDC = test mode
0 VDC (or open circuit) = normal mode

b. Two wire highlighted range contour line select.

| Selected Contour line | RNG 4/RNG 6 Wire | RNG 5/RNG 6 Wire |
|---|---|---|
| (1) Range 3 | 0 | 0 |
| (2) Range 4 | +28 VDC | 0 |
| (3) Range 5 | 0 | +28 VDC |
| (4) Range 6 | +28 VDC | +28 VDC |

c. TC-CAL off/on - 0 VDC shall result in TC confidence markers being displayed.

3.4.8.5 Inputs from radar altimeter.

3.4.8.5.1 Altitude above terrain. Radar altitude information shall be provided directly from the radar altimeter system; the signal characteristics are shown below:

| | |
|---|---|
| a. Operating voltage range | +1.0 to +21.0 VDC |
| b. Altitude range | 0 to 5000 feet |
| c. Maximum rate of change | 3000 ft/sec |
| d. Scale factor, 0-400 ft. | 20 mv/ft. |
| e. Voltage level, 0 ft. | +1.0 VDC |
| f. Voltage level, 400 ft. | +9.0 VDC |
| g. Voltage level, 5000 ft. | +21.0 VDC |
| h. Maximum voltage | +25 VDC |
| i. Minimum RDC input impedance | 100K |

3.4.9 Outputs from the radar data converter. The following outputs from the RDC are tabulated in Table V.

3.4.9.1 Outputs to the ADI. The outputs specified in 3.4.9.1.3 will not be used if the aircraft or the ADI have not been modified.

3.4.9.1.1 RDC video. The RDC shall provide video to the ADI. The signal shall be supplied over a 93 ohm line. The voltage levels are:

| | |
|---|---|
| a. Bright | 1.6 ± 0.2 volts |
| b. Black: | 0 + 0.2, -0.0 volts |



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TABLE V. Output signal characteristics.

| Output number | Signal name | Function | Signal type | Voltage range | Zin-input impedance | RDC connector pin | Signal destination |
|---|---|---|---|---|---|---|---|
| 1 | RDC video | TC display | video | 1.6V pp | To drive 90 ohms | J2-13 (Coax) | ADI |
| 2 | Altimeter validity | Light RA indicator | DC | +28 VDC = on open=off | To drive 120 milliamp lamp load | J1-16 | ADI |
| 3 | RDC validity | TC warning light | DC | +28 VDC = on open=off | To drive 200 milliamp lamp load | J1-15 | ADI thru search radar |
| 4 | TC CAL azimuth gate | Enable TC calibration | Pulse | 0 ± 0.5V 11 ± 3V | To drive 1K min. | J1-17 | Search radar |
| 5 | TC CAL azimuth gate return | Signal return | - | - | - | J1-18 | Search radar |
| 6 | TC CAL azimuth gate shield | Shield termination | - | - | - | J1-19 | Search radar |
| 7 | +15 VDC | Test set elevation pot excitation | DC | +15 VDC | To supply 5 ma | J1-3 | RDC test set |
| 8 | -15 VDC | Test set elevation pot excitation | DC | -15 VDC | To supply 5 ma | J2-19 | RDC test set |





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TABLE V. Output signal characteristics - Continued

| Output number | Signal name | Function | Signal type | Voltage range | Zin-input impedance | RDC connector pin | Signal destination |
|---|---|---|---|---|---|---|---|
| 9 | FLIR/Ext mode over-ride | Override | TTL | 0/5V,0V= override | TTL | J3-M | ADI |
| 10 | TAD Status | Status | DC | 0/5V,0V= TAD | - | J3-L | ADI |
| 11 | FLIR/Ext override return | Ground ref | - | - | - | J3-P | ADI |





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3.4.9.1.2 Altimeter validity. A +28 VDC discrete signal shall be supplied to the ADI to light the radar altitude reliable indicator light whenever the altitude above terrain signal (see 3.4.8.5.1) is greater than +21.5 ± 0.5 VDC or less than +1.4 ± 0.5 VDC.

3.4.9.1.3 FLIR/External mode override. A TTL compatible output shall be supplied to the ADI to indicate a tactical altitude director alarm condition. A TTL low shall indicate an alarm and a TTL high shall indicate a safe condition.

3.4.9.1.4 TAD status. An output line shall be supplied to the ADI to report the TAD status of the aircraft and the ADI. If the ADI TAD input is open (aircraft or ADI unmodified) the voltage of this output shall be +5V and if the ADI TAD input is grounded, the output shall be 0V.

3.4.9.2 Outputs to search radar.

3.4.9.2.1 RDC validity. A +28 VDC discrete signal shall be supplied to the search radar whenever the RDC automatic built-in-test circuits, or failure detection circuitry detects a failure.

3.4.9.2.2 TC CAL azimuth gate. An antenna azimuth gate pulse corresponding to an antenna azimuth angle of +18.0 ± 0.5 degrees to +26.5 ± 0.5 degrees to the right of the aircraft centerline, shall be supplied to the search radar. This pulse shall be a positive 11 ± 3V pulse in sync with the antenna azimuth scan signal. The TC CAL azimuth gate shall be supplied to the search radar only when the TC CAL off/on signal is on (0 VDC) and the test mode on/off signal is off (0 VDC).

3.5 Detail design requirements.

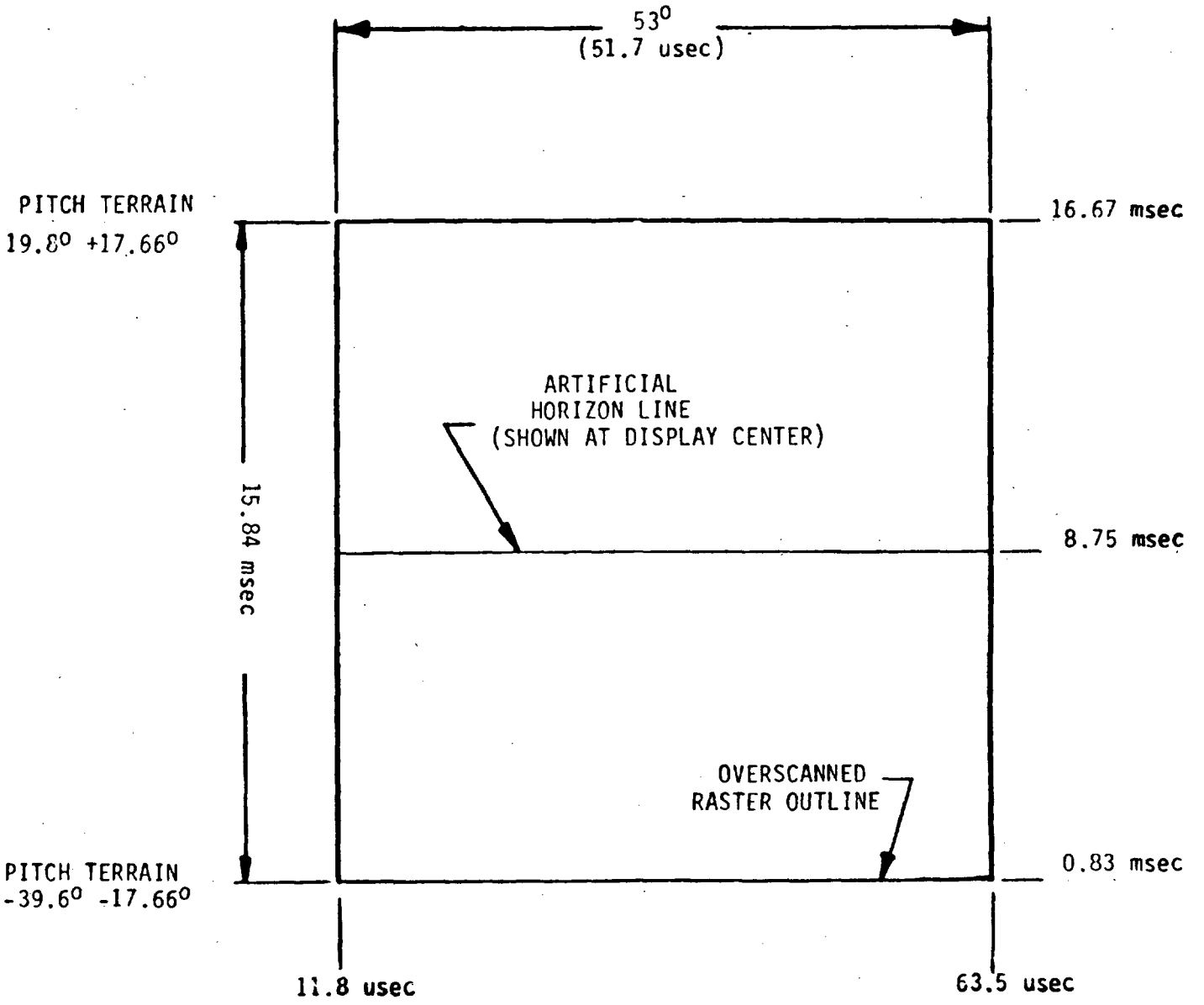
3.5.1 Display format. The terrain clearance presentation shall be displayed on a raster as shown in Figure 5. The radar video shall correspond to 35.3 terrain elevation degrees in vertical by 53 azimuth degrees in horizontal. Above the raster center (0 degree), the elevation (pitch) shall have a vertical scaling factor of 17.66 degrees of terrain elevation per 19.8 degrees of pitch elevation. Below the raster center (0 degree), the elevation (pitch) shall have a vertical scaling factor of 17.66 degrees of terrain elevation per 39.6 degrees of pitch elevation. Movement of the artificial horizon line shall be determined by scaled elevation (pitch).

3.5.1.1 Raster. The RDC shall provide RDC video compatible with a 525 line raster with 2:1 interlace. The scan shall be from bottom to top and from left to right.

3.5.1.2 Frame rate. The frame rate of the RDC video shall be 30 Hz. The field rate shall be 60 Hz.



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NOTE: 1. Times shown are with respect to leading edge of either the horizontal or vertical blanking signal
2. Field rate = 60 Hz.
3. Horizontal blanking = 11.8 microseconds

FIGURE 5. TC raster with artificial horizon line at 0°.



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3.5.2 Shaded gray terrain clearance display. The RDC shall generate video of a shaded gray terrain clearance presentation. This presentation shall consist of discrete shades of gray of eight contour ranges, varying from very bright for the closest range to a dark gray at the farthest range. The darkest range line shall be readily discernable from the black artificial horizon line. A continuous black line consisting of pixels one raster line vertically and one azimuth bin (0.255 degrees) horizontally shall be generated at the range contour boundaries. The output voltage levels for each range contour shall be as specified in Table VI.

TABLE VI. Range boundaries.

| Range Contour | Output Voltage Amplitude (below horizon line) |
|---|---|
| 1 | 1.60 ± 0.020 VDC |
| 2 | 1.50 ± 0.020 VDC |
| 3 | 1.40 ± 0.020 VDC |
| 4 | 1.25 ± 0.025 VDC |
| 5 | 1.10 ± 0.025 VDC |
| 6 | 0.90 ± 0.025 VDC |
| 7 | 0.75 ± 0.025 VDC |
| 8 | 0.55 ± 0.025 VDC |

3.5.2.1 Amplitude scaling. The output voltage amplitude for range contour video above the horizon line shall increase by 0.1 volts.

3.5.2.2 Memory failure. The algorithm used to generate the eight range contours shall continue to function if a failure of half of the memory occurs, except that the video shall contain only range one through four.

3.5.2.3 Distribution of range contour lines. The processed TC video shall be sampled over a 106.67 µsec (8.5 nautical miles) interval. The nominal distribution of range samples shall be as specified in Table VII.

TABLE VII. Distribution of range contour lines.

| Range Contour line | Sample interval range (delta) in nautical miles | | Sample interval in microseconds |
|---|---|---|---|
| 1 | 0.25 - 0.50 | (0.25) | 4.40 - 7.79 |
| 2 | 0.50 - 0.75 | (0.25) | 7.79 - 10.88 |
| 3 | 0.75 - 1.00 | (0.25) | 10.88 - 13.97 |
| 4 | 1.00 - 1.50 | (0.50) | 13.97 - 20.15 |
| 5 | 1.50 - 2.00 | (0.50) | 20.15 - 26.33 |
| 6 | 2.00 - 2.50 | (0.50) | 26.33 - 32.51 |
| 7 | 2.50 - 4.50 | (2.00) | 32.51 - 57.23 |
| 8 | 4.50 - 8.50 | (4.00) | 57.23 -106.67 |

Note: The sample interval reflects a 1.3 microsecond lag of the search radar TC video with respect to TCP to RDC trigger.

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3.5.2.4 Sample interval accuracy. The leading edge of each range gate shall occur within ± 5 percent of its specified location. The trailing edge of each gate shall be coincident with the leading edge of its succeeding gate. The total range sampling period shall be 106.67 ± 5.4 microseconds.

3.5.2.5 Azimuth resolution. There shall be 208 increments of azimuth information displayed on the 53 degree overscanned raster.

3.5.2.6 Incremental distribution of azimuth elements. The 208 azimuth increments shall be linearly distributed with each azimuth increment spanning 0.255 degrees.

3.5.2.7 Azimuth element averaging. As the antenna sweeps through each 0.25 degree of azimuth, or display bin, incoming radar data shall be averaged with the other radar inputs falling within this same azimuth bin to form the initial data elements. Only the first eight inputs shall be used; typically only six radar inputs occur within each bin. These initial elements shall then be averaged using a moving window algorithm which shall perform as a function of the range highlighting selection. When either range three or six is selected for highlighting, a moving window of 16 elements shall be used and when either range four or five is selected, a moving window of 30 elements shall be used. In each case, the previous 15 or 29 elements will have been already averaged and shall now be averaged with the current initial element to create the current average element.

3.5.2.8 Synchronous clamping level. The RDC shall clamp the zero degree horizon of the processed TC video from the search radar to the zero degree (horizon) reference.

3.5.3 Highlight bars. The RDC shall generate highlight bars which allow the pilot to discriminate between range contours. The highlight bars shall always be present on range one and range seven, and shall be individually selectable for ranges three, four, five, and six. The highlight bars shall be black to contrast with the range contour and shall move to the left or right as a function of the azimuth angle input. The highlight bars shall be spaced as shown in Figure 6.

3.5.4 Black altitude curtain. The RDC shall generate a black altitude curtain which indicates ground clearance determined from the altitude above terrain input. The black altitude curtain shall be black and shall blank all processed TC video below it. The vertical distance between the top of the black altitude curtain and the artificial horizon line shall be determined with a scale factor of 0.534 milliseconds per volt as related to the vertical sync input. The black altitude curtain shall be blanked if the altitude above terrain input is greater than 21.5 volts or less than 1.4 volts.

3.5.5 TC calibration markers. When the TC CAL input is "on", the RDC shall generate two black calibration markers at the right side of the raster (+18.0 to +26.5 degrees). One marker shall be developed from ranges one, two, and three. The second marker shall be developed from ranges six, seven, and eight. Each marker shall start at the range with the lowest elevation and end at the range with the highest elevation within the range group. A minimum thickness of one raster line shall be generated. The valid marker positions are 0 degree and -6 degrees with respect to the artificial horizon line.



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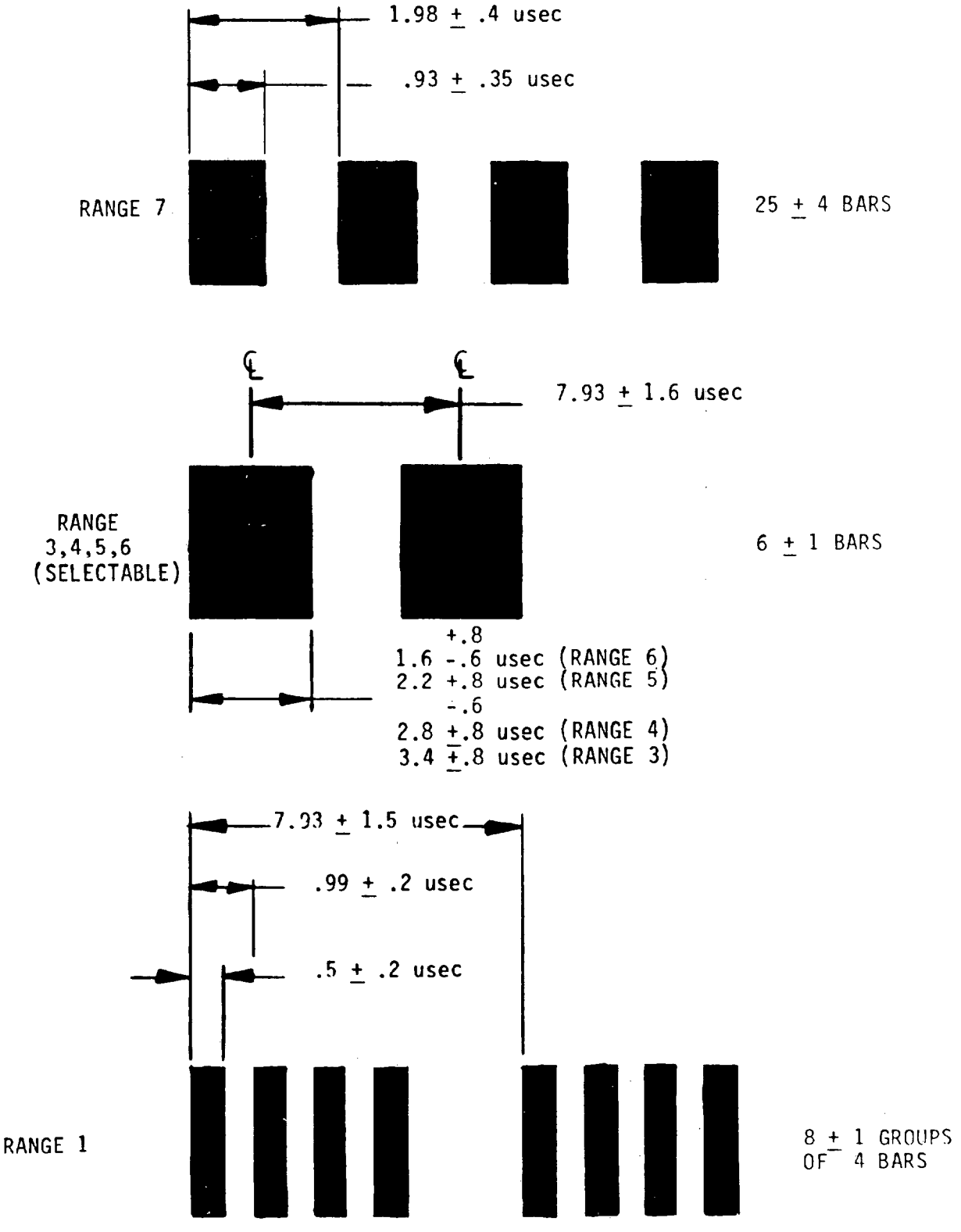



FIGURE 6. Highlight bar characteristics.



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3.5.6 Artificial horizon line. The RDC shall generate an artificial horizon line which shall provide elevation angle information with a static position accuracy of seven raster lines.

3.5.6.1 Dynamic range. The artificial horizon line shall be driven by the elevation angle between the limits of ± 30 degrees.

3.5.6.2 Characteristics. The artificial horizon line shall be a black line at 0 degrees elevation and two raster lines wide.

3.5.6.3 Relative positional accuracy. The relative positional accuracy between the TC presentation and the artificial horizon line shall be not greater than ±7 raster lines.

3.5.7 Sky video. Areas in the TC raster which will appear above the artificial horizon line and do not contain any range contour information shall be the sky shade of +1.20 volts.

3.5.8 Ground video. Areas in the TC raster which will appear below the artificial horizon line and represent ground beyond 8.5 nautical miles shall be the ground shade of +1.10 volts.

3.5.9 Tactical altitude director. The Tactical Altitude Director (TAD) system shall provide an indication of close ground proximity and director type steering about the lateral axis of the aircraft under specific operating conditions dependent on aircraft and ADI modification. With full system implementation the TAD symbol shall be operational under all aircraft conditions with the exception of: gear down and locked, aircraft roll greater than 60 degrees, or flight path angle (FPA) greater than +45 degrees. In the event the aircraft or ADI are not modified for TAD, the symbol shall only be displayed with TC video. In this case, there shall be no provision for landing cutout in the TC mode. The TAD system shall analyze terrain conditions from the airplane present position out to 1 to 2.5 nautical miles, dependent on the range highlight selection.

3.5.9.1 TAD select. The RDC shall contain a jumper wire which shall blank the TAD symbol from the TC video. When the jumper wire is removed the symbol shall be enabled.

3.5.9.2 TAD mechanization. The TAD algorithm shall cause the TAD symbol to be linearly positioned between, or at, its stowed position and its alarm position dependent on specific parameters. These parameters include the encountered terrain elevation and range, the aircraft pitch, velocity vector, pseudo energy state, radar altitude, altitude above anticipated terrain, radar antenna pitch destabilization, and the selected clearance mode. The value of the upper and lower limits displayed by the symbol shall define the TAD safety zone which shall be optimized to meet safety and mission requirements. The upper altitude of the safety zone shall be indicated by symbol movement from the stowed position. Penetration of the safety zone lower altitude limit shall be indicated by the symbol rising to a maximum vertical height approximately equal to the electronic center of the display, and flashing at a 3.0 ± 1 Hz rate. The upper and lower limits of the TAD safety zone will be dependent on range highlight selection and aircraft FPA. Both primary and secondary algorithm functions shall independently determine the position of the aircraft relative to the TAD safety zone. The TAD symbol shall be positioned by the algorithm function indicating the lowest altitude condition.



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3.5.9.2.1 Primary function. The primary algorithm function shall determine TAD symbol position as a function of terrain elevation angle at range X (ELx), and range Y (ELy) in 25 elements about the impact point in a ± 3 degree azimuth window as follows:

$$TADI \leq ELx' - FPA \times F(A) + Beta \times F(TA) \leq TADA$$

where: TADI = symbol initial position
ELx' = antenna pitch destabilization
FPA = flight path angle
F(A) = altitude function
Beta = aircraft energy state
F(TA)= terrain anticipation function
TADA = symbol alarm position

ELx and ELy are variable range elevations dependent on the range highlight selection as defined in Table VIII.

TABLE VIII. Variable range selection.

| Highlighted Range | Memory Status | TC Mode Selected | Range used for ELx and ELy | |
|---|---|---|---|---|
| | | | X | Y |
| 3 | ok | on | 2 | 3 |
| 3 | fail | on | 2 | 3 |
| 4 | ok | on | 3 | 4 |
| 4 | fail | on | 3 | 4 |
| 5 | ok | on | 5 | 6 |
| 5 | fail | on | 3 | 4 |
| 6 | ok | on | 1 | 3 |
| 6 | fail | on | 1 | 3 |
| 3 | ok | off | 2 | 3 |
| 3 | fail | off | 2 | 3 |
| 4 | ok | off | 3 | 4 |
| 4 | fail | off | 3 | 4 |
| 5 | ok | off | 5 | 6 |
| 5 | fail | off | 3 | 4 |
| 6 | ok | off | * | * |
| 6 | fail | off | * | * |

*Range bin information shall not be used in this mode.



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3.5.9.2.2 Secondary functions. The second algorithm functions shall be independent of the primary and each other, and shall effect symbol position as a function of both selected mode and FPA. The secondary algorithm functions shall be as follows:

a. TSP1 = F(EL1)

b. TSP3 = F(EL3)

c. TSPR = F(RDRALT)

where: TSP = TAD symbol position
F(EL1) = range 1 function
F(EL3) = range 3 function
F(RDRALT) = radar altitude function

If the landing gear signal is open/0 volts DC for less than 20 seconds, the above secondary algorithm functions shall be set to zero unless FPA is less than 2 degrees. The radar altitude and the elevation angles of various range bins dependent on FPA and selected mode shall be as shown in Table IX.

3.5.9.3 Symbol description. The symbol appearance shall change as a factor of radar input and ADI mode as shown in Figure 7. The RDC shall blank the range 1 highlight bars around the TAD symbol. In the ADI Contact Analog mode bright ADI symbols may overlay portions of the displayed TAD symbol.

3.5.9.4 Alarm condition. When the TAD symbol position reaches the electronic center of the display an alarm condition shall be indicated and the symbol shall flash at a rate of 3.0 ± 1.0 Hz.

3.5.9.5 Symbol movement. Vertical movement of the top of the symbol shall be determined by the algorithm. The symbol shall be in its stowed position when the algorithm solution indicates a completely safe condition. As the algorithm solution indicates increasing proximity to the terrain, the top of the symbol shall move toward the electronic center of the display.

3.5.9.6 Modes. The algorithm shall have four calculation modes. Mode selection shall correspond to the coded range contour as specified below:

| Mode | Coded range contour |
|---|---|
| a. Extremely low (50 feet) | 6 |
| b. Low (125 feet) | 3 |
| c. High (360 feet) | 4 |
| d. High (360 feet over extremely rough terrain) | 5 |

3.5.9.6.1 Independant mode. When the ADI or the aircraft have not been modified to provide the impact point horizontal and impact point vertical positions, the algorithm shall assume a vertical impact point of -2.5 degrees from the aircraft ADL. When the ADI or the aircraft have not been modified to provide landing gear status, the algorithm shall assume the landing gear is retracted.



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TABLE IX. Radar altitude and elevation angles.

| HIGHLIGHTED RANGE | FPA | SYMBOL POSITION | RADAR ALTITUDE | $EL_1$ | $EL_3$ | $EL_x$ | $EL_y$ |
|---|---|---|---|---|---|---|---|
| 6* | 0° | MAX | 50 ± 5 | - | - | -0.80 ± 0.25 | -0.40 ± 0.25 |
| | | MIN | 610 ± 25 | - | - | -11.4 ± 1.0 | -5.7 ± 0.5 |
| | -20° | MAX | 1300 ± 25 | - | - | -25.9 ± 1.5 | -12.7 ± 1.0 |
| | | MIN | 1850 ± 40 | - | - | -32.0 ± 2.0 | -25.4 ± 2.0 |
| 3 | 0° | MAX | 125 ± 10 | -2.35 ± 0.30 | - | -1.52 ±0.30 | -1.14 ±0.3 |
| | | MIN | 750 ± 25 | -14.2 ± 1.0 | - | -9.4 ± 1.0 | -7.1 ± 1.0 |
| | -20° | MAX | 1325 ± 25 | -26.2 ± 2.0 | - | -17.3 ± 1.5 | -12.9 ± 1.0 |
| | | MIN | 1950 ± 40 | - | - | -29.3 ± 2.0 | -21.4 ± 1.5 |
| 4 | 0° | MAX | 300 ± 15 | -5.66 ± 0.50 | - | -3.38 ± 0.50 | -2.25 ± 0.30 |
| | | MIN | 1055 ± 25 | -20.0 ± 1.5 | - | -10.0 ± 1.0 | -6.6 ± 0.5 |
| | -20° | MAX | 1350 ± 25 | -26.1 ± 2.0 | - | -12.9 ± 1.0 | -8.5 ± 1.0 |
| | | MIN | 2100 ± 40 | - | - | -23.6 ± 1.5 | -15.4 ± 1.0 |
| 5 | 0° | MAX | 300 ± 15 | -5.66 ± 0.50 | -2.83 ± 0.3 | -1.70 ± 0.30 | -1.36 ± 0.3 |
| | | MIN | 1910 ± 40 | - | -18.2 ± 1.5 | -9.0 ± 1.0 | -7.2 ± 1.0 |
| | -20° | MAX | 1350 ± 25 | -26.1 ± 2.0 | -12.8 ± 1.0 | -6.8 ± 1.0 | -5.4 ± 0.5 |
| | | MIN | 2950 ± 60 | - | -28.1 ± 2.0 | -16.9 ± 1.5 | -13.5 ± 1.0 |

* In contact analog, this mode uses radar altitude only.

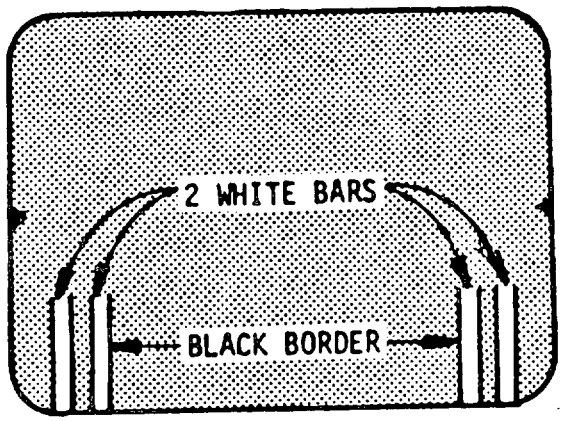




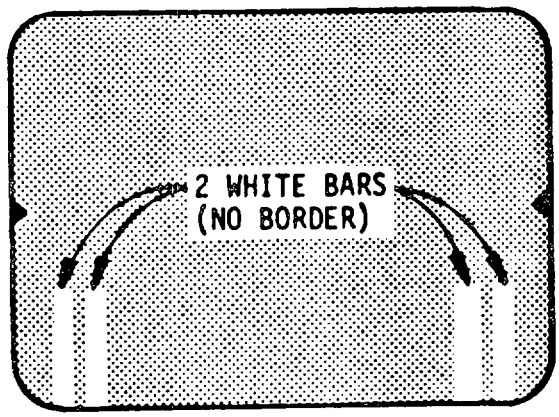
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SEARCH RADAR WITH/WITHOUT RADAR ALTIMETER

TERRAIN CLEARANCE

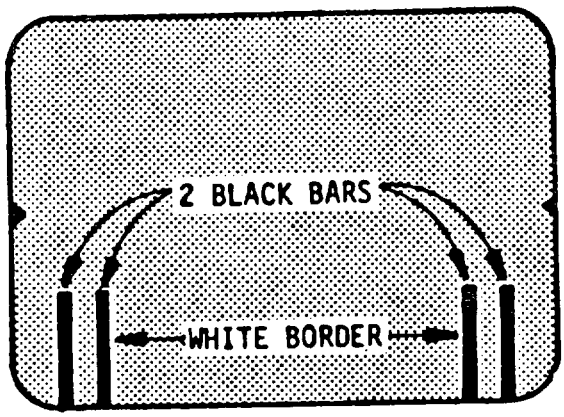



CONTACT ANALOG




RADAR ALTIMETER ONLY

TERRAIN CLEARANCE




CONTACT ANALOG

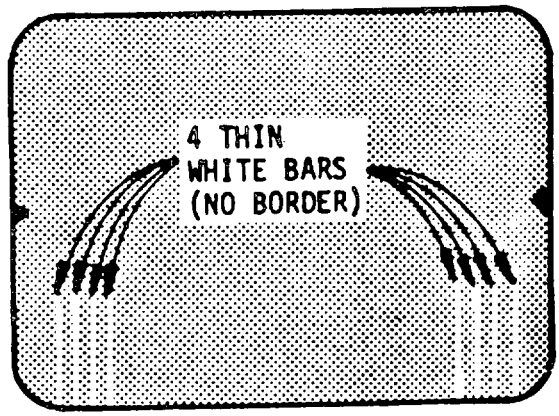



FIGURE 7. TAD symbol appearance.



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3.5.9.6.2 Automatic mode change. Automatic algorithm mode changes shall occur for steep dive angles.

3.5.9.7 Stowed position. When the algorithm indicates a completely safe condition, the TAD symbol shall be in a static position within 4.3 ± 0.3 milliseconds after the ADI vertical sync signal.

3.5.9.8 Radar monitor. The TAD symbol shall flash while in the stowed position under the following conditions.

a. The azimuth scan of the search radar is outside of a 3 degree window about the impact point.

b. The pulse repetition frequency of the search radar is less than 800 Hz.

3.5.9.9 ADI mode switchover. When the algorithm indicates an alarm condition, the FLIR/Ext override output shall be a TTL compatible low.

3.5.9.10 Memory loss. In the event that half of the RDC's memory fails, the algorithm shall continue to calculate the ground proximity.

3.5.9.11 Blanking. The TAD symbol shall be blanked under the following conditions:

a. Landing gear down and aircraft and ADI have been modified for TAD
b. RDC validity output voltage 28 VDC and altimeter output voltage 28 VDC
c. RDC failure

3.6 Environmental requirements. Unless otherwise specified herein, the RDC shall operate as specified under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment.

3.6.1 Standard conditions. The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the RDC:

| | | |
|---|---|---|
| a. | Temperature | Room ambient (25 ± 10°C) |
| b. | Altitude | Normal ground |
| c. | Vibration | None |
| d. | Humidity | Room ambient to 90 percent relative humidity |
| e. | Voltage | 115 ± 1.0 VAC, 3 phase, 400 ± 20 Hz and 27.5 ± 0.5 VDC |

3.6.2 Temperature altitude. The RDC shall operate as specified when subjected to the temperature altitude conditions for Class 2 equipment as specified in MIL-E-5400, except the maximum altitude shall be 55,000 feet.

3.6.3 Vibration. The RDC shall operate as specified when subjected to the random vibration levels of MIL-E-5400.



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3.6.4 Shock. The RDC shall operate as specified when subjected to the equipment shock requirements of MIL-E-5400. In addition the RDC or dummy load shall withstand crash safety testing in accordance with the requirements of MIL-E-5400.

3.6.5 Thermal shock. The RDC shall operate as specified when subjected to sudden changes in temperature.

3.6.6 Humidity. The RDC shall withstand exposure to humidities up to 100 percent.

3.6.7 Salt atmosphere. The RDC shall operate as specified when exposed to salt-sea atmosphere.

3.6.8 Explosion. The RDC shall not cause an ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere.

3.6.9 Sand and dust. The RDC shall withstand exposure to sand and dust particles.

3.6.10 Fungus. The RDC shall withstand exposure to fungus growth.

3.7 Electromagnetic interference. The RDC shall control electromagnetic emission and susceptibility as specified in MIL-STD-461.

3.8 Nomenclature. Nomenclature, serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (4.5)

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in paragraph 3.6.1.



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4.4 First article inspection. First article tests shall be conducted by the contractor on two RDC's representative of the production RDC's to be supplied under the contract. The Government inspector and the procuring activity shall be advised 10 days in advance of when tests are to be conducted so that a Government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.4.1 First article test procedures. A first article test procedure shall be made available to the procuring activity for acceptance. (See 6.2.2)

4.4.2 First article test data. The contractor shall make available all data collected in conducting these tests to the procuring activity for review.

4.4.3 Scope of tests. First article tests shall consist of the following tests:

a. Power characteristics test (see 4.6.1)
b. Acoustic noise (see 4.6.2)
c. Individual tests (see 4.6.3)
d. Environmental tests (see 4.6.4)
e. Electromagnetic interference (see 4.6.5)
f. Maintainability demonstration tests (see 4.6.6)
g. Reliability qualification tests (see 4.6.7)

4.4.4 Test sequence. The test sequence shall be acceptable to the procuring activity.

4.5 Quality conformance inspection. Quality conformance tests shall be conducted on each production RDC. All inspection and testing shall be under supervision of the government inspector. The contractor shall include test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product.

4.5.1 Scope of tests. Quality conformance testing shall consist of the following tests:

a. Individual tests (see 4.6.3)
b. Manufacturing screening test (see 4.6.8)
c. Reliability acceptance tests (see 4.6.9)

4.6 Test methods.

4.6.1 Power characteristics test. The RDC shall be tested to demonstrate the requirements of MIL-STD-704 with the following additions.

4.6.1.1 Power consumption test. The RDC shall be operated at maximum power consumption and shall be not greater than the limits specified in 3.4.1.1.

4.6.1.2 Under-voltage test. Reduce the line voltage from standard condition to zero and then back to standard conditions over a period of five minutes. There shall be no damage to any components as a result of this test.



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4.6.2 Acoustic noise. The RDC shall be tested in accordance with MIL-T-5422.

4.6.3 Individual tests. Each RDC submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, and operational adequacy. As a minimum, each RDC accepted shall have passed the following tests:

a. Examination of product
b. Operational test

4.6.3.1 Examination of product. Each RDC shall be examined carefully to determine that the material and workmanship requirements have been met, and that the physical characteristics of the RDC agree with the outline dimensions specified.

4.6.3.2 Operational test. Each RDC shall be operated until the RDC temperature stabilizes, prior to conducting the operational test. The test shall verify the operational performance requirements specified in Section 3.

4.6.3.3 Test procedure. An acceptance test procedure shall be made available to the procuring activity for acceptance. (See 6.2.2)

4.6.4 Environmental. Prior to and after each environmental test, operational tests shall be performed as specified in 4.6.3.2 to verify the compliance with the requirements specified in Section 3.

4.6.4.1 Temperature and altitude. The RDC shall be tested in accordance with MIL-T-5422, except the upper altitude limit shall be 55,000 feet.

4.6.4.2 Vibration. The RDC shall be tested in accordance with MIL-STD-810.

The category shall be b.2 and $W_o$ shall be 0.0126 $G^2$/Hz.

4.6.4.3 Shock. The RDC shall be tested in accordance with MIL-T-5422.

4.6.4.4 Thermal shock. The RDC shall be tested in accordance with MIL-T-5422.

4.6.4.5 Humidity. The RDC shall be tested in accordance with MIL-T-5422.

4.6.4.6 Salt fog test. The RDC shall be tested in accordance with MIL-T-5422.

4.6.4.7 Explosion. The RDC shall be tested in accordance with MIL-T-5422.

4.6.4.8 Sand and dust. The RDC shall be tested in accordance with MIL-T-5422.

4.6.4.9 Fungus. The RDC shall be tested in accordance with MIL-T-5422.

4.6.5 Electromagnetic interference. The RDC shall be tested in accordance with MIL-STD-462.

4.6.6 Maintainability demonstration tests. A maintainability demonstration test shall be conducted in accordance with MIL-STD-470 and MIL-STD-471, limited to intermediate level maintenance only and shall verify RDC compliance with the specified corrective maintenance time specified in 3.3.6.1.3. In addition, the tests shall demonstrate the BIT function and self test features are capable of



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fault detection and isolation to a WRA, the repair is accomplished through WRA removal and replacement, and that repair verification is accomplished exercising the stated equipment BIT and self test features.

4.6.7 Reliability qualification test. Two RDC's shall be tested in accordance with MIL-STD-781. Test plan XIVC shall be used. The temperature shall be cycled from -54°C to +71°C as delineated in Figure 9, except the exact time of each event shall be determined by thermal survey. Each RDC shall be normally mounted and subjected to random vibration as delineated in Figure B-4 of MIL-STD-781, except $W_0$ shall be 0.0024. The duration of vibration shall be at least 30 minutes during each hour of "on time". Failures shall be classified in accordance with MIL-STD-2074.

4.6.8 Manufacturing screening test. Each production RDC shall be subjected to random vibration in order to stimulate the early occurrence of failures due to manufacturing defects in parts and workmanship. The equipment shall be operational and monitored for proper performance during random vibration cycle.

4.6.8.1 Test duration. The duration of vibration shall be 10 minutes.

4.6.8.2 Test axis. The RDC shall be hardmounted in the axis perpendicular to the printed circuit boards.

4.6.8.3 Test spectrum. The screening test shall consist of random vibration in accordance with Figure 9.

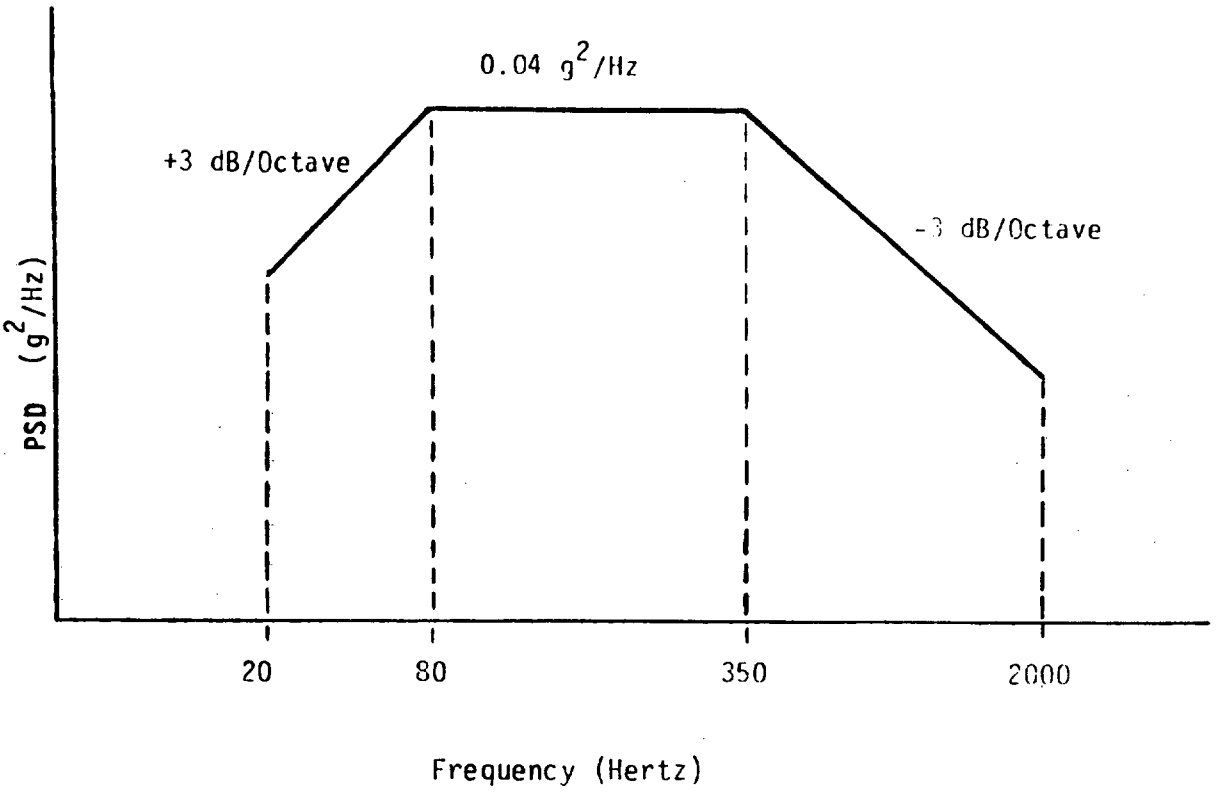



FIGURE 8. Random vibration spectrum.



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4.6.9 Reliability acceptance test. The RDC shall be subjected to a Reliability Acceptance Test (RAT) in accordance with MIL-STD-781.

4.6.9.1 All equipment test. Each production RDC, except those submitted for the reliability qualification testing, shall be tested in accordance with the all equipment production reliability acceptance test of MIL-STD-781. Test plan XVIIIC shall be used with the accept-reject criteria derived from test plan IVC, shown in Figure C-13 of MIL-STD-781. The test period for each production RDC shall be 100 failure free hours.

4.6.9.2 Test profile. RAT shall consist of repeated eight hour cycles as shown in Figure 9. The random vibration shall be in accordance with MIL-STD-781, except $W_0$ shall be 0.0024.

4.6.9.3 Failure classification. Failures shall be defined and classified in accordance with MIL-STD-781 and MIL-STD-2074.

4.6.9.4 Equipment burn-in. Prior to the 100 hour failure free test, the RDC shall be subjected to a burn-in period of not less than 3 cycles using the test profile shown in Figure 9.

4.6.9.5 Test MTBF. The upper test MTBF ($\theta_0$) shall be 1200 hours and the lower test MTBF ($\theta_1$) shall be 600 hours.

4.6.9.6 Accept-reject criteria. Accept-reject criteria shall be in accordance with MIL-STD-781. At any time that the totals of test hours and test failures show a reject situation, the procuring activity shall be notified immediately. The procuring activity reserves the right to stop the acceptance of RDC's at any time that a reject situation exists.

4.6.9.7 Test details. The test details such as the length of test cycle, temperature and vibration profile, performance characteristics to be measured and any special failure criteria shall be part of the test procedure made available to the procuring activity for acceptance (see 6.2.2).

4.7 Test procedures. The procedures used for conducting first article test and quality conformance tests shall be prepared by the contractor and made available to the procuring activity for review and acceptance. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used for preparation of



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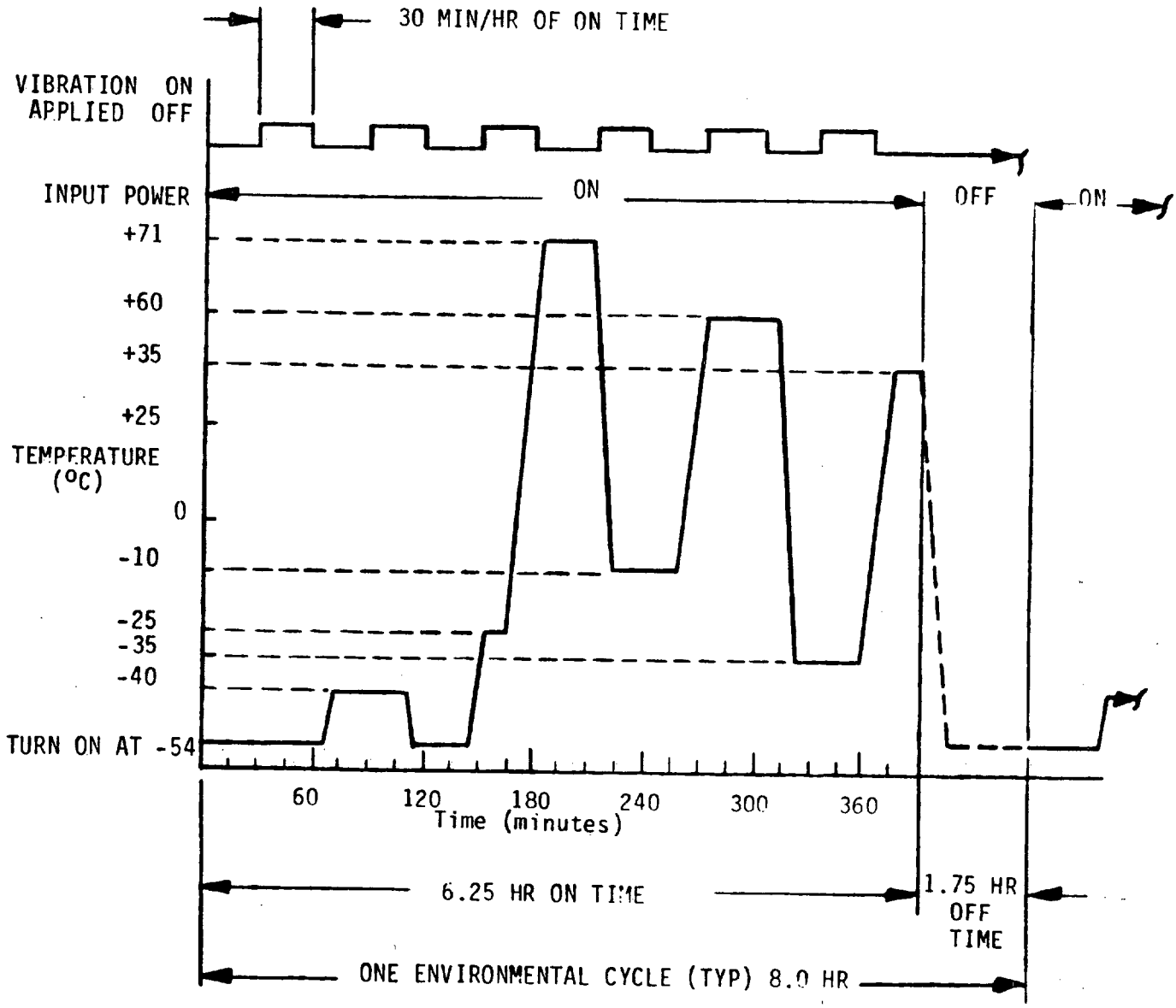



Figure 9. RAT environmental cycle.



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test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional test, when deemed necessary.

5. PACKAGING

5.1 General. All major units and parts of the RDC shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the RDC is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. The RDC is intended to quantize and store radar video and generate the video signal necessary to provide a terrain clearance display on the ADI. The RDC will accomplish this by utilizing inputs from the peripheral sensor equipments (radar, radar altimeter, inertial subsystem, etc.).

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition requirements should specify the following:

a. Title number and date of this specification.
b. Selection of applicable levels of packaging and packing.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

| Paragraph No. | Data requirement title | Applicable DID No. |
|---|---|---|
| 3.3 | Design Data Report | DI-R-24039A |
| 4.4.1, 4.7 | First Article Test Procedure | UDI-T-21347 |
| 4.6.3.3 | Acceptance Test Procedure | UDI-T-21347 |
| 4.6.9 | Reliability Acceptance Test Procedure | DI-R-7035 |
| 4.7 | Quality Conformance Test Procedure | UDI-T-21347 |
| 4.6.7 | Reliability Qualification Test Procedure | DI-R-7035 |
| 4.6.7 | Reliability Qualification Test Report | DI-R-7034 |
| 4.6.6 | Plan Maintainability Demostration | DI-R-2129 |
| 4.6.6 | Maintainability Demonstration Report | DI-R-7113 |



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(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item(s) will be tested and should be a first article sample. The first article should consist of two units. The Contracting Officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval.

6.4 Associated equipment. The RDC should operate and be compatible with the following associated equipment which is part of the A6 aircraft to the extent that their inputs and outputs are specified in Tables I and V. The associated equipment should not be supplied as part of the RDC.

| | | |
|---|---|---|
| a. | Analog display indicator | IP-722( )/AVA-1 |
| b. | Search radar | AN/APQ-92, -148, -156 |
| c. | Inertial subsystem | AN/ASN-31, -92 |
| d. | Radar altimeter | AN/APN-141, -194 |
| e. | Pilots control panel | 128AV66804, 66826 |

Preparing Activity
Navy-AS
(Project No. 5841-N029)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-85638(AS) | Military Specification For Converter, Radar Data |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

**5. PROBLEM AREAS**

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

**6. REMARKS**

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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